Chapter 6.

§6.1

A — $n \times n$ matrix

$$AX = \lambda X \quad (X \neq 0)$$

λ ↓ eigenvalue; X ↳ corresponding eigenvector.

i) Characteristic polynomial $\underline{\det(A-\lambda I)} = 0 \Leftrightarrow \lambda$ is an eigenvalue.

↳ degree n. in λ.

ii) For each λ, solve $(A-\lambda I)X = 0$ to find corresponding eigenvectors.

Theorem: i) $\det A = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$

ii) $\lambda_1 + \lambda_2 + \cdots + \lambda_n = \underline{\text{Trace}(A)} = a_{11} + a_{22} + \cdots + a_{nn}$.

Imaginary eigenvalues/eigenvectors — appear in "Conjugate Pairs".

§6.2. <u>diagonization of Matrix</u>

Theorem: $S^{-1}AS = \Lambda$ or $A = S\Lambda S^{-1}$ if A has n linearly independent eigenvectors.

where $S = [x_1\ x_2 \cdots x_n] \rightarrow$ eigenvector matrix

$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \rightarrow$ eigenvalue matrix

$$Ax_i = \lambda x_i, \quad i = 1, 2, \cdots, n.$$

Theorem: i) Eigenvectors corresponding to distinct eigenvalues are linearly independent.

ii) An $n \times n$ matrix that has n different eigenvalues must be diagonizable.

<u>Power of A</u>: $A^k = S\begin{bmatrix} \lambda_1^k & & & \\ & \lambda_2^k & & \\ & & \ddots & \\ & & & \lambda_n^k \end{bmatrix}S^{-1}$ if $A = S\Lambda S^{-1}$

§6.4.

symmetric matrix $A = A^T$

Theorem: i) The eigenvalues of a real symmetric matrix are real.

ii) Eigenvectors of a real symmetric matrix are always perpendicular.

Theorem: Every symmetric matrix A has a complete set of orthogonal eigenvectors:

$$A = S\Lambda S^{-1} \longrightarrow A = Q\Lambda Q^T$$

§6.5.

Definition of Positive definiteness of a matrix A.

$$(\text{symmetric}) + (\lambda_i > 0,\ i=1,\cdots,n) \Rightarrow A \text{ positive definite.}$$

$$+ (\lambda_i \geq 0,\ i=1,\cdots,n) \Rightarrow A \text{ semidefinite.}$$

Theorem: $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite $\Leftrightarrow$ $a > 0$ and $ac - b^2 > 0$.

Theorem: The following statements are equivalent for a symmetric matrix A:

1) A is positive definte
2). All n eigenvalues are positive.
3) $X^TAX > 0$ if $X \neq 0$
4) All n pivots are positive
5) All n upper left determinants are positive.

Cholesky Factorization.

$A = LDL^T$ when A is symmetric.

$\longrightarrow$ $A = (L\sqrt{D})(L\sqrt{D})^T$ Since D is diagonal with positive numbers on the diagonal.